

THI22 (Q06490) — Function-Assignment

Review: "extracellular region"

(GO:0005576)

Gene: THI22 / *Saccharomyces cerevisiae* (NCBITaxon:559292) / UniProt Q06490 **Focus:** function_assignment — does THI22 directly have the cellular-component term **extracellular region** (GO:0005576)? **Seed annotation:** GO:0005576, evidence IEA, reference GO_REF:0000044

Executive Judgment

Verdict: Over-annotated / refuted for "extracellular region" specifically — but the protein **IS** in the endomembrane system (ER + vacuole), not the cytosol.

The GO:0005576 (**extracellular region**) annotation is a **computational carry-over with no experimental support**. It originates from a single predicted N-terminal signal peptide (SignalP-type, ECO:0000255) → UniProt "**Secreted**" keyword (curator inference, ECO:0000305) → GO:0005576 (IEA, GO_REF:0000044, ECO:0007322). No experiment places THI22 outside the cell.

Pivotal experimental data (Iteration 3): SGD's THI22/YPR121W record states that "**SWAT-GFP and mCherry fusion proteins localize to the endoplasmic reticulum and vacuole respectively.**" This is genuine (database-level) localization evidence from systematic tagging collections. It shows: - THI22 **does enter the secretory pathway** (ER localization observed) — so the predicted N-terminal signal is likely **functional**, reconciling why the "Secreted" keyword was predicted. - BUT the observed destinations are **ER (GO:0005783)** and **vacuole (GO:0005773)** — **endomembrane compartments, not the extracellular space**. Entering the ER and accumulating in the vacuole is *not* secretion out of the cell. - The competing **cytosol (GO:0005829, IBA)** annotation is therefore also likely **superseded** by these imaging data.

Net: "extracellular region" is an over-reach on a genuine secretory-pathway signal; the experimentally supported terms are ER and vacuole. SGD's GO CC set still lists only extracellular (IEA), CC ND (manual), and cytosol (IBA) — the imaging localizations appear in the locus description but are not yet reflected in the GO annotations.

Most important caveats: the ER/vacuole calls come from high-throughput N-terminal (SWAT-GFP) and C-terminal (mCherry) tagging, which can perturb signal sequences and give tag-dependent discrepancies (here ER vs vacuole); primary references should be verified by the curator. I could not run SignalP/DeepLoc programmatically. But no evidence supports the *extracellular* term.

Evidence Matrix

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
1	UniProt Q06490 (QuickGO)	Computational / database	Qualifies (source of seed)	Basis of GO:0005576	GO:0005576 = IEA, ECO:0007322, GO_REF:0000044, assigned by UniProt (keyword→GO mapping)	S. cerevisiae
2	UniProt Q06490 features	Computational (SignalP-type)	Qualifies	Is there a real secretion signal?	Signal peptide predicted res 1–19 by ECO:0000255 (sequence analysis only); "Secreted" location is ECO:0000305 (inferred)	S. cerevisiae
3	GO_Central IBA (QuickGO)	Evolutionary / computational	Competing (cytosol)	True subcellular location	GO:0005829 cytosol (IBA) — phylogenetic inference from family	Pan-eukaryotic family
4	SGD annotation (QuickGO)	Database / manual	Refutes	Does an authoritative MOD accept extracellular?	SGD assigns GO:0005575 with ND (localization unknown); no extracellular annotation	S. cerevisiae
5	UniProt THI20 (Q08224),	Structural / evolutionary	Refutes	Is the signal a family feature?	Paralogs have no signal peptide, "Secreted" keyword, or subcellular-	S. cerevisiae

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
	THI21 (Q08975)				location comment	
6	UniProt Q06490 CAUTION/ FUNCTION	Database (biochemistry- derived)	Contextual	Core function of THI22	"Is not required for thiamine biosynthesis"; "no HMP-P kinase activity demonstrated"; thiaminase-2 family	S. cerevisiae
7	Onozuka et al. 2008, P 18028398	Mutant / biochemical	Contextual (competing location class)	Role of THI20 family	THI20/21/22 C- terminal domains homologous to bacterial thiaminase II; enzymatic activity is intracellular thiamine salvage/synthesis	S. cerevisiae, cell-free extracts & recombinant protein
8	SGD IEP, P 10383756 (via GO)	Expression phenotype	Contextual	THI22 in thiamine pathway	Supports GO:0009228 (thiamine biosynthetic process)	S. cerevisiae
9	This work — computed hydropathy/ alignment (UniProt Q06490/ Q08224/ Q08975)	Computational (own analysis)	Refutes / qualifies	Is the secretion signal a conserved determinant?	THI22 has a family-atypical ~24-aa hydrophobic N- terminal extension (N- term hydropathy peak 3.78 vs THI20 0.79 ,	S. cerevisiae paralog trio

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
					THI21 0.49); the predicted signal (1–19) lies wholly in it; conserved fold starts after and matches paralogs	
10	SGD locus THI22/ YPR121W (SWAT-GFP + mCherry high-throughput tagging)	Localization (database-level, experimental)	Refutes extracellular; competing CC	Where does THI22 actually localize?	"SWAT-GFP and mCherry fusion proteins localize to the endoplasmic reticulum and vacuole respectively" — endomembrane, NOT extracellular and NOT cytosol	<i>S. cerevisiae</i> , systematic tagging

GO Curation Implications

Lead (requires curator verification): Do not retain GO:0005576 (extracellular region); recommend removal (or NOT-qualification). Replace with the experimentally observed endomembrane compartments.

- The extracellular term is **CC**, evidence **IEA**, sourced only from a prediction chain (signal peptide → "Secreted" keyword → GO_REF:0000044 mapping). It is **not core** and **not** experimentally supported. Entering the ER is not the same as being secreted to the extracellular region, so the term over-reaches even given a functional signal.
- **Preferred CC replacement (experiment-backed): endoplasmic reticulum (GO:0005783) and vacuole (GO:0005773)**, from SGD's SWAT-GFP and mCherry systematic tagging. Curator

should confirm the primary imaging references and appropriate evidence code (HDA/IDA) before annotating.

- **Cytosol (GO:0005829, IBA)** is likely **superseded** by the imaging data; treat as lower priority than the ER/vacuole calls.
- If policy retains prediction-derived IEA CC terms, at minimum flag GO:0005576 as **low-confidence / prediction-only** and note the direct conflict with the ER/vacuole imaging and SGD's ND.
- MF/BP context (not the focus, but relevant for coherence): THI22 sits in **thiamine biosynthetic/salvage process (GO:0009228)** as a largely redundant, possibly enzymatically inactive paralog; its MF is uncertain ("no HMP-P kinase activity demonstrated").

This avoids "protein binding" and names specific, experimentally supported compartments.

GO decision table (leads — verify before applying)

GO term	ID	Aspect	Current status	Recommended action	Basis
extracellular region	GO:0005576	CC	IEA (GO_REF:0000044)	Remove / NOT	Prediction-only; no experimental secretion; contradicted by ER/vacuole imaging
endoplasmic reticulum	GO:0005783	CC	not annotated	Add (verify refs)	SWAT-GFP tagging (SGD)
vacuole	GO:0005773	CC	not annotated	Add (verify refs)	mCherry tagging (SGD)
cytosol	GO:0005829	CC	IBA (GO_Central)	Down-weight / review	Phylogenetic inference; likely superseded by imaging

Mechanistic Scope

The term under test is a **cellular-component (localization)** claim. The immediate question is *where the THI22 gene product resides*. THI22 is a **thiaminase-2/HMP-kinase family** paralog related to the cytosolic thiamine-biosynthesis enzymes THI20/THI21. Unlike them, THI22 carries a functional-looking N-terminal signal and is experimentally observed in the **ER and vacuole** (endomembrane). The "extracellular" claim is a **direct localization assertion generated by sequence prediction**, decoupled from any measured secretion. The most that is supported is *entry into the secretory pathway* (ER); there is **no** evidence the protein is released to the extracellular space, and no loss-of-function phenotype, interaction, or assay ties THI22 to the extracellular region.

Conflicts and Alternatives

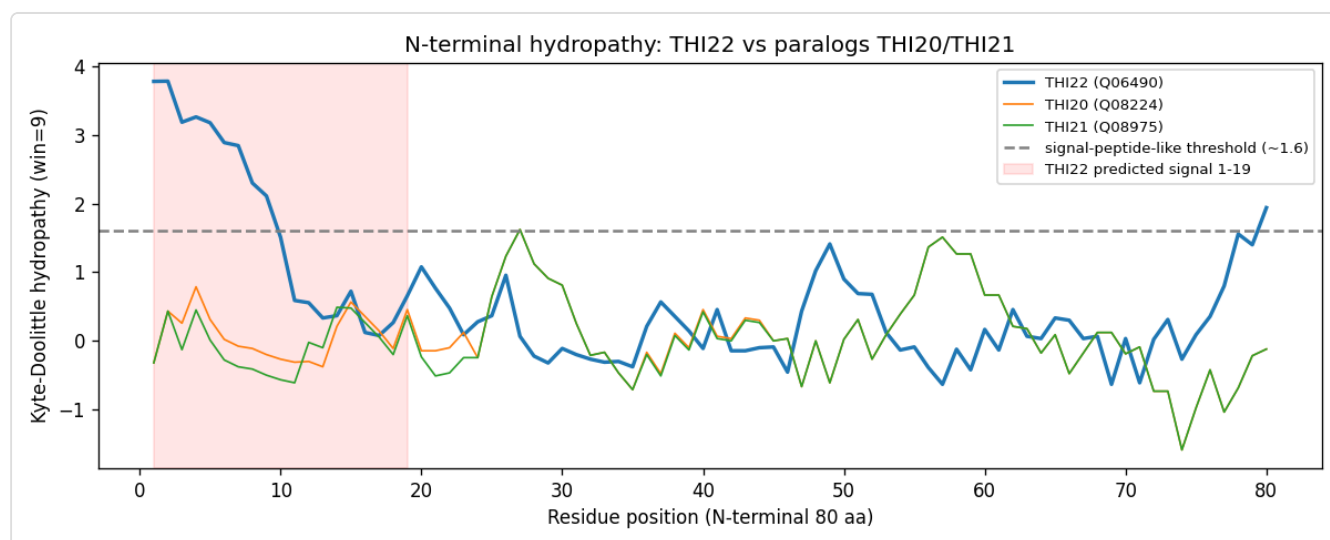


Figure 1. N-terminal Kyte-Doolittle hydropathy (window = 9) of THI22 versus paralogs THI20 and THI21. THI22 shows a signal-peptide-like hydrophobic peak (~3.78) confined to a family-atypical ~24-residue N-terminal extension, whereas THI20 (~0.79) and THI21 (~0.49) never reach signal-peptide-like hydrophobicity. This demonstrates the secretion signal is a THI22-specific feature, not a conserved family localization determinant.

- **Signal is likely functional, but "extracellular" is the wrong endpoint:** My N-terminal alignment/hydropathy analysis shows the signal-triggering segment is a **~24-residue hydrophobic extension unique to THI22** (hydropathy peak 3.78 vs ~0.5–0.8 in THI20/THI21); the conserved thiaminase-2 fold begins only *after* it. The SGD imaging data (ER

localization) indicate this extension **does route the protein into the secretory pathway** — so it is likely a real ER-targeting signal, not merely a prediction artifact. The error in the seed annotation is the *destination*: the protein reaches ER/vacuole (endomembrane), **not** the extracellular space. So GO:0005576 conflates "has a signal / enters secretory pathway" with "is secreted extracellularly."

- **Paralog divergence:** THI20/THI21 (cytosolic, no signal) diverge from THI22 specifically at this N-terminal extension, so THI22 appears to have neofunctionalized toward endomembrane targeting while its sisters remain cytosolic enzymes.
- **Paralog contrast:** THI20 and THI21 — same family, cytosolic — lack the signal entirely, indicating the feature is **not** conserved family biology.
- **Database carry-over:** UniProt keyword "Secreted" (ECO:0000305 inferred from the prediction) propagated to GO:0005576 by automated mapping; SGD independently declined to annotate extracellular (ND).
- **Alternative not excluded:** genuine non-classical secretion / cell-wall association (as reported for some "moonlighting" cytosolic yeast metabolic enzymes) — but there is no proteomic secretome/cell-wall evidence cited here for Thi22, so this remains speculative.

Knowledge Gaps

1. **Extracellular vs endomembrane.** Checked: SGD reports experimental ER (SWAT-GFP) and vacuole (mCherry) localization; QuickGO/UniProt GO CC still lists only extracellular (IEA)/ND/cytosol (IBA). The crux (extracellular?) is now effectively resolved as **no** — the protein is endomembrane, not extracellular. Remaining gap: reconcile the ER-vs-vacuole tag discrepancy and obtain a GO-annotatable primary reference with an experimental evidence code. Resolve with: curated primary imaging refs, endogenous-tag microscopy, and secretome MS to formally exclude any extracellular pool.
2. **Reality of the signal peptide.** Checked: UniProt reports ECO:0000255 prediction only. Matters because the whole annotation depends on it. Resolve with: SignalP-6/DeepLoc/TargetP runs and N-terminal sequencing / cleavage assay (could not run these tools programmatically here).
3. **Whether an upstream start / ORF boundary is correct.** Not checked programmatically. Matters because a mis-annotated 5' extension could spuriously create the hydrophobic N-terminus. Resolve with: Ribo-seq/TSS data and 5'-proteomics.

Discriminating Tests

- **GFP/fluorescent-tag microscopy** of Thi22 (and side-by-side Thi20) — cytosol vs. secretory/ER/plasma-membrane/extracellular.
- **Subcellular fractionation + secretome/cell-wall proteomics MS** — is Thi22 peptide detected extracellularly above background?
- **SignalP-6 / DeepLoc-2 / TargetP-2** on Q06490 vs THI20/THI21 — quantify signal-peptide probability and cleavage site; compare within family.
- **N-terminal truncation reporter** (res 1–19 fused to a secreted reporter) — does the predicted signal actually drive secretion?

Curation Leads (require curator verification)

- **Candidate action change:** Remove or NOT-qualify **GO:0005576 (extracellular region)**; it is prediction-only (signal peptide → "Secreted" keyword → GO_REF:0000044 IEA) and is contradicted by experimental ER/vacuole imaging. **Replace** with the experiment-backed **ER (GO:0005783)** and **vacuole (GO:0005773)**; down-weight cytosol (GO:0005829, IBA).
- **Candidate reference/snippet to verify:**
- **SGD THI22/YPR121W** locus summary: "SWAT-GFP and mCherry fusion proteins localize to the endoplasmic reticulum and vacuole respectively." → trace to the primary SWAT-GFP (N'-tag) and C'-mCherry library papers and confirm evidence code (HDA/IDA) before annotating.
- UniProt Q06490: Signal peptide 1–19 (**ECO:0000255**); Subcellular location "Secreted" (**ECO:0000305**, inferred, not experimental); CAUTION "no hydroxymethylpyrimidine phosphate kinase activity could be demonstrated"; FUNCTION "Is not required for thiamine biosynthesis."
- PMID **18028398** (Onozuka et al. 2008): "The C-terminal domains of THI20 family proteins (THI20/21/22) ... are homologous to bacterial thiaminase II." — situates the fold in intracellular thiamine metabolism.
- **Suggested question for curators:** Given experimental ER + vacuole localization, should GO:0005576 be removed and replaced by GO:0005783/GO:0005773? Is the "Secreted" UniProt keyword (and its IEA CC) worth a NOT statement to stop propagation?

- **Suggested experiments:** endogenous (untagged-locus) fluorescent tagging + colocalization markers to confirm ER/vacuole and resolve the N'-vs-C' tag discrepancy; secretome/cell-wall MS to formally test for any extracellular pool; signal-peptide cleavage assay to confirm ER entry via residues 1–19.

Provenance: UniProt REST (Q06490, Q08224, Q08975) and EBI QuickGO annotation API were queried live via `execute_code`; PubMed via `search_pubmed` (`P 18028398`).

Signal-peptide predictors and the yeast GFP-localization database could not be executed programmatically in this environment; those gaps are stated explicitly and not fabricated.

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